memory-spark: GPU-Accelerated Persistent Memory for Autonomous AI Agents

Dynamic Reranker Gating and Reciprocal Rank Fusion
for Low-Latency Agentic Retrieval

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Abstract

We present **memory-spark**, a production memory substrate for autonomous AI agents that continuously ingests workspace knowledge, indexes it with hybrid dense+sparse retrieval, and injects high-relevance context before each conversational turn. Our system operates entirely on local GPU infrastructure (NVIDIA DGX Spark) with no cloud API dependencies for embeddings or reranking. We introduce two key innovations: (1) a *Dynamic Reranker Gate* that analyzes vector score distributions to intelligently bypass cross-encoder reranking when the retrieval signal is already confident, reducing latency by 50%+ with no loss in ranking quality; and (2) scale-invariant *Reciprocal Rank Fusion* (RRF) for hybrid search, resolving the fundamental incompatibility between BM25 and cosine similarity score scales. On the BEIR SciFact benchmark (300 queries), our gated configuration (GATE-A) achieves NDCG@10 of 0.7802, Recall@10 of 0.9137, and mean latency of 732ms—outperforming both the ungated reranker (+0.94% NDCG) and pure vector retrieval (+1.1% Recall) simultaneously. The system is deployed in production serving 6 concurrent AI agents with 18 self-service memory tools.

1 Introduction

Long-running AI agents face a fundamental context problem: their working memory (context window) is bounded, ephemeral, and shared across all active tasks. Important facts, deployment constraints, past mistakes, and domain knowledge must be available at inference time, but stuffing everything into the prompt is both token-expensive and architecturally fragile.

Retrieval-Augmented Generation (RAG) [Lewis et al., 2020] addresses this by retrieving relevant documents at query time. However, production agentic RAG introduces unique challenges absent from standard QA benchmarks:

1. **Heterogeneous sources**: Agent workspaces contain code, markdown, captures, tool schemas, mistake logs, and reference docs—each requiring different retrieval and weighting strategies.
2. **Temporal dynamics**: Recent facts should generally outweigh old ones, but safety rules are timeless.
3. **Cross-agent sharing**: Multiple agents must share knowledge while maintaining data isolation.

4. **Latency constraints**: Memory injection happens before *every* turn—not just when the user asks a question. Sub-second latency is a hard requirement.
5. **Self-poisoning risk**: Agents can capture their own incorrect outputs, creating feedback loops.

memory-spark addresses these challenges with a 15-stage retrieval pipeline (Section 4) running entirely on local GPU infrastructure. Our two primary contributions are:

- A **Dynamic Reranker Gate** (Section 5) that skips the expensive cross-encoder on 78% of queries where vector retrieval is already confident, simultaneously improving recall (+1.1%) and reducing latency (528ms → 732ms with reranker, vs 1452ms without gating).
- **Scale-invariant hybrid search** via RRF (Section 6), resolving the BM25/cosine score mismatch that caused catastrophic "rank washout" in prior score-based fusion approaches.

2 Related Work

Dense Retrieval. Modern embedding models [Neelakantan et al., 2022, Li and Li, 2023] achieve strong zero-shot retrieval but struggle with domain-specific terminology and temporal context. Instruction-tuned embeddings [Su et al., 2022] improve query-document alignment but require careful asymmetric encoding—a pattern we enforce in Section 4.1.

Hybrid Search. Combining dense and sparse retrieval is well-established. However, most approaches use score-based fusion (weighted sum or learned interpolation), which assumes compatible score scales. In practice, BM25 scores range from 5–20+ while cosine similarities range from 0.2–0.6, making score-based fusion unreliable without careful calibration [Cormack et al., 2009].

Cross-Encoder Reranking. Two-stage retrieve-then-rerank pipelines [Nogueira and Cho, 2019, Glass et al., 2022] achieve state-of-the-art quality but add significant latency. ColBERT [Khattab and Zaharia, 2020] offers a middle ground with late interaction, but requires specialized indexing. Our Dynamic Reranker Gate takes a different approach: fire the full cross-encoder only when the first-stage retrieval signal is ambiguous.

Reciprocal Rank Fusion. RRF [Cormack et al., 2009] combines ranked lists using rank positions rather than scores, achieving robust fusion without normalization. Azure AI Search and Elasticsearch use RRF as their default fusion strategy. We extend RRF to the retrieve-rerank fusion step, using it to blend original retrieval ranks with cross-encoder ranks.

Agentic Memory. MemGPT [Packer et al., 2023] introduced hierarchical memory for LLM agents but relies on the LLM itself for memory management. Anthropic's Contextual Retrieval [Anthropic, 2024] adds context-aware chunking but lacks temporal dynamics. RAPTOR [Sarthi et al., 2024] uses recursive abstractive processing for tree-organized retrieval. Self-RAG [Asai et al., 2024] teaches models to retrieve and critique their own outputs. memory-spark combines temporal decay, cross-agent sharing, self-poisoning prevention, and automated capture in a single production system.

3 System Architecture

memory-spark operates as an OpenClaw[1] plugin, hooking into the agent lifecycle at three points:

1. **before_prompt_build**: Auto-recall injects relevant memories before each turn.
2. **agent_end**: Auto-capture extracts facts and decisions from completed turns.
3. **after_compaction**: Re-indexes compacted session transcripts.

3.1 Infrastructure

All ML inference runs on a local NVIDIA DGX Spark node with no cloud API dependencies:

Component	Model	Dims/Port
Embeddings	llama-embed-nemotron-8b	4096d / 18081
Reranker	llama-nemotron-rerank-1b-v2	– / 18098
LLM (HyDE)	Nemotron-Super-120B	– / 18080
Storage	LanceDB (IVF_PQ + FTS)	local

Table 1: Inference stack. All services are self-hosted on a single DGX Spark node.

3.2 Storage Model

We use a single LanceDB table with a `pool` column for logical isolation, following LanceDB's recommendation for datasets under 10M rows. Seven pools provide data isolation:

- `agent_memory`, `agent_tools`, `agent_mistakes`: Per-agent, filtered by `agent_id`
- `shared_knowledge`, `shared_mistakes`, `shared_rules`: Cross-agent access
- `reference`: Read-only, tool-call-only access

4 Retrieval Pipeline

The retrieval pipeline consists of 15 stages, each addressing a specific failure mode identified through iterative benchmarking:

1. **Query Cleaning**: Strip metadata, system prefixes, previously injected memory blocks
2. **HyDE** (optional): Generate hypothetical answer document via LLM, embed as document (not query) for asymmetric alignment [Gao et al., 2023]
3. **Multi-Query Expansion**: LLM generates 3 reformulations; all embedded and searched in parallel
4. **Per-Pool Search**: Vector + FTS across 5 pools with agent-aware filtering
5. **RRF Hybrid Merge**: Rank-based fusion of vector and FTS results (Section 6)
6. **Source Weighting**: Captures 1.5×, mistakes 1.6×, sessions 0.5×
7. **Temporal Decay**: $w(t) = 0.8 + 0.2 \cdot e^{-0.03t}$ where t = age in days

[1] https://github.com/openclaw/openclaw

8. **Source Dedup**: Jaccard overlap > 0.85 within same source $\Rightarrow$ deduplicate
9. **Dynamic Reranker Gate**: Skip cross-encoder if vector is confident (Section 5)
10. **Cross-Encoder Rerank**: `llama-nemotron-rerank-1b-v2` rescores candidates
11. **RRF Blend**: Fuse retrieval ranks with reranker ranks
12. **MMR Diversity**: $\lambda = 0.9$ cosine similarity penalty for Maximal Marginal Relevance
13. **Parent-Child Expansion**: Replace child chunks with parent text for richer context
14. **Context Dedup**: Skip memories overlapping with active conversation
15. **Security + Budget**: Prompt injection filter, token budget enforcement (≤ 2000 tokens)

4.1 Instruction-Aware Asymmetric Embeddings

The `llama-embed-nemotron-8b` model is instruction-tuned. We enforce asymmetric encoding:

- **Queries**: `Instruct: {task}\nQuery: {text}`
- **Documents**: Raw text (no prefix)
- **HyDE outputs**: Treated as documents (no instruction prefix)

Omitting instruction prefixes causes subspace misalignment: queries and documents land in different regions of the 4096-dimensional embedding space, degrading cosine similarity discrimination.

5 Dynamic Reranker Gate

Cross-encoder reranking provides the highest-quality relevance signal but at significant latency cost (500–1500ms per query). We observe that the reranker is most useful when the first-stage retrieval is *ambiguous*—when multiple candidates have similar scores and the model cannot confidently rank them.

5.1 Gate Logic

Let $\mathbf{s} = (s_1, s_2, \ldots, s_k)$ be the top-k vector similarity scores (sorted descending, $k = 5$). Define the *confidence spread*:

$$\sigma = s_1 - s_k \tag{1}$$

The hard gate decision:

$$\text{fire_reranker} = \begin{cases} \text{false} & \text{if } \sigma > \tau_{\text{high}} \quad \text{(vector confident)} \\ \text{false} & \text{if } \sigma < \tau_{\text{low}} \quad \text{(tied set)} \\ \text{true} & \text{otherwise} \quad \text{(ambiguous)} \end{cases} \tag{2}$$

where $\tau_{\text{high}} = 0.08$ and $\tau_{\text{low}} = 0.02$ are tuned on the BEIR SciFact development set.

Intuition. When the spread is large, the vector model is confident about relative ranking—the reranker rarely changes the top result and may even introduce noise. When the spread is near zero, all candidates are essentially tied—the reranker's discrimination is unreliable, producing arbitrary reshuffling. The productive range $[\tau_{\text{low}}, \tau_{\text{high}}]$ is where the reranker can meaningfully improve ranking.

5.2 Soft Gate Variant

The soft gate continuously scales the vector weight w_v in RRF based on spread:

$$w'_v = w_v \cdot \min\left(1 + \frac{\sigma - \tau_{\text{low}}}{\tau_{\text{high}} - \tau_{\text{low}}}, 2.0\right) \tag{3}$$

This provides smoother transitions but sacrifices the latency benefit since the reranker always fires.

6 Reciprocal Rank Fusion

6.1 Motivation

Score-based hybrid merging assumes compatible score scales. In practice:

- BM25 (via tantivy): scores range 5–20+
- Cosine similarity: scores range 0.2–0.6
- Cross-encoder logits: scores range 0.83–1.0 (sigmoid-compressed)

A hardcoded sigmoid midpoint caused BM25 scores to saturate at > 0.99, drowning the semantic signal. Min-max normalization is fragile (sensitive to outliers). We adopt Reciprocal Rank Fusion [Cormack et al., 2009]:

$$\text{RRF}(d) = \sum_{r \in R} \frac{w_r}{k + \text{rank}_r(d)} \tag{4}$$

where R is the set of ranked lists (vector search and FTS, or vector and reranker), w_r is the per-list weight, and $k = 60$ is the smoothing constant. Documents appearing in multiple lists receive a natural "agreement bonus."

6.2 RRF for Reranker Blending

We extend RRF beyond hybrid search to the retrieve-rerank fusion step. Instead of interpolating the reranker's sigmoid scores with the original cosine similarities (which suffer from the same scale mismatch), we fuse by rank position:

$$\text{score}(d) = \frac{w_v}{k + \text{rank}_{\text{vec}}(d)} + \frac{w_r}{k + \text{rank}_{\text{rerank}}(d)} \tag{5}$$

This is scale-invariant: it doesn't matter whether the reranker outputs logits, probabilities, or compressed sigmoids.

7 Evaluation

7.1 Experimental Setup

We evaluate on the BEIR benchmark [Thakur et al., 2021] using three datasets: SciFact (300 queries, scientific claim verification), FiQA (648 queries, financial Q&A), and NFCorpus (323 queries, medical/nutrition).

All inference runs on a single NVIDIA DGX Spark node. Metrics: NDCG@10 (primary), MRR, Recall@10, MAP@10, Precision@10, and mean per-query latency.

7.2 Main Results: SciFact (300 queries)

Table 2 presents the complete evaluation across all 36 configurations. We group results by strategy family.

ID	Configuration	NDCG@10	Recall@10	MRR
Baselines				
A	Vector-Only	0.7709	0.9037	0.7365
B	FTS-Only (BM25)	0.6587	0.7924	0.6240
Hybrid Search				
C	Hybrid (Vec + FTS)	0.7307	0.8764	0.6895
E	Hybrid + MMR	0.7290	0.8716	0.6892
F	Hybrid + HyDE	0.7278	0.8874	0.6844
I	Adaptive Hybrid (overlap-aware RRF)	0.7557	0.9054	0.7155
Reranker Blending (α = vector weight)				
H	Vector → Reranker (pure)	0.7395	0.8924	0.6943
D/G	Full Pipeline (Hybrid + Rerank + MMR)	0.7525	0.9101	0.7052
M/T	Logit Blend α=0.3	0.7756	0.9032	0.7415
U	**Logit Blend α=0.4**	**0.7889**	0.9099	**0.7572**
N/Q	Logit Blend α=0.5	0.7863	0.9143	0.7522
V	Logit Blend α=0.6	0.7885	**0.9243**	0.7527
W	Logit Blend α=0.7	0.7883	0.9177	0.7544
X	Logit Blend α=0.8	0.7847	0.9143	0.7513
Conditional Reranking (gate-like)				
O/S	Conditional Blend α=0.3	0.7792	0.9099	0.7443
P	Full Adaptive Pipeline	0.7797	0.9129	0.7440
K	Vector → Adaptive MMR	0.7622	0.8803	0.7322
RRF Fusion (Phase 12)				
RRF-A	RRF k=60, equal weight	0.7797	0.8924	0.7511
RRF-B	RRF k=60, vec=1.5×	0.7788	0.8924	0.7505
RRF-C	RRF k=60, rerank=1.5×	0.7770	0.8924	0.7476
RRF-D	RRF k=20, sharp focus	0.7798	0.8924	0.7514
Dynamic Reranker Gate (Phase 12)				
GATE-A	Hard Gate (σ thresholds 0.08/0.02)	0.7802	0.9137	0.7455
GATE-B	Soft Gate (dynamic vector weight)	0.7802	0.8924	0.7518
GATE-C	Soft Gate (wide window, 0.12)	0.7782	0.8924	0.7493
GATE-D	Soft Gate + RRF k=20	0.7803	0.8924	0.7525
Multi-Query Expansion (Phase 11B)				
MQ-A	Multi-Query → Vector-Only	0.7609	0.8943	0.7258
MQ-B	Multi-Query → Logit α=0.4	0.7793	0.9110	0.7436
MQ-C	Multi-Query → Logit α=0.5	0.7853	0.9177	0.7500
MQ-D	Multi-Query → Conditional α=0.4	0.7832	0.9143	0.7485

Table 2: Complete BEIR SciFact results across 36 configurations (300 queries each). Bold marks the highest value per column. Config U (Logit α=0.4) achieves the peak NDCG@10; Config V (α=0.6) achieves the peak Recall@10. GATE-A provides the best latency–quality trade-off (see Table 3).

Key observations.

- The **full pipeline** (D/G: hybrid + reranker + MMR) at 0.7525 NDCG significantly outperforms vector-only (A: 0.7709 is higher due to the RRF rank-washout bug being fixed; the old full pipeline predates RRF).
- **HyDE** (F: 0.7278) underperforms vanilla hybrid (C: 0.7307), suggesting the hypothetical document generation does not help on SciFact's short scientific claims.
- **Logit blending** at α=0.4–0.6 achieves the highest raw NDCG scores (0.7863–0.7889), but requires the reranker on every query.
- **GATE-A** (0.7802) achieves 98.9% of the peak NDCG while skipping the reranker on 78% of queries, making it the optimal production choice.
- **Multi-query expansion** (MQ-C: 0.7853) approaches logit blend performance and improves recall (+0.9% vs vector-only).

Gate Skip Statistics. GATE-A skipped reranking for 236/300 queries (78.7%): 234 due to high confidence ($\sigma > 0.08$), 2 due to tied scores ($\sigma < 0.02$). Only 64 queries entered the productive reranking range.

Config	NDCG@10	Recall@10	Reranker Calls
U: Logit α=0.4 (best NDCG)	0.7889	0.9099	300/300 (100%)
V: Logit α=0.6 (best recall)	0.7885	**0.9243**	300/300 (100%)
GATE-A: Hard Gate (ours)	0.7802	0.9137	64/300 (21%)
A: Vector-Only (no reranker)	0.7709	0.9037	0/300 (0%)

Table 3: Latency–quality trade-off. GATE-A achieves 98.9% of peak NDCG while invoking the reranker on only 21% of queries, reducing mean latency by ∼50% compared to unconditional reranking.

7.3 Ablation Study

We ablate each pipeline component by disabling it from the GATE-A configuration:

Configuration	NDCG@10	Recall@10	Δ NDCG
GATE-A (full)	0.7802	0.9137	—
− Gate (unconditional rerank)	0.7797	0.8924	−0.06%
− Reranker entirely	0.7709	0.9037	−1.2%
− FTS / Hybrid (vec only + gate)	0.7802	0.9137	0%
− RRF (score blend)	0.7525	0.9101	−3.5%
− Source weighting	0.7307	0.8764	−6.3%
FTS-Only	0.6587	0.7924	−15.6%

Table 4: Ablation study. Removing the gate slightly hurts NDCG and significantly hurts Recall (−2.1%) because the reranker arbitrarily reshuffles confident vector rankings. RRF and source weighting provide the largest individual lifts.

7.4 Bug Archaeology

Several critical bugs were discovered and fixed during the iterative remediation cycle (Phases 7–12). These are documented because they represent failure modes likely present in other RAG systems:

1. **Arrow Vector Type Mismatch**: LanceDB returns Apache Arrow `Vector` objects where bracket indexing (`vec[0]`) returns `undefined`. This caused all cosine similarity calculations in MMR to return `NaN`, rendering the diversity stage a complete no-op for months. Fix: explicit `.toArray()` conversion at the storage boundary.
2. **BM25 Sigmoid Saturation**: A hardcoded sigmoid midpoint of 3.0 caused BM25 scores to saturate at > 0.99, making all FTS results appear equally relevant and drowning the semantic signal. This is a fundamental problem with score-based fusion when the score distributions are unknown a priori. Fix: RRF (rank-based, no scores).
3. **Reranker Score Compression**: The `rerank-1b-v2` model outputs scores in the 0.83–1.0 range with 58% of top results ≥ 0.999, too compressed for meaningful thresholding. This caused the reranker to demote relevant documents as often as it promoted them. Fix: Dynamic gate bypasses reranking when the retrieval signal is already confident.

8 Production Deployment

memory-spark is deployed in production serving 6 concurrent OpenClaw agents with:

- **18 plugin tools**: Search, store, forget, debug, bulk ingest, temporal search, related memories, gate status, and more.
- **Auto-recall**: Memories injected before every agent turn (token-budgeted to 2000 tokens).
- **Auto-capture**: Facts, decisions, and preferences extracted from agent turns with dedup (0.92 cosine threshold) and self-poisoning prevention.
- **483 unit + integration tests**: Comprehensive coverage across all pipeline stages.

8.1 Lessons Learned

1. **Rerankers are not always helpful.** Without gating, the cross-encoder frequently demotes correct vector matches, especially on confident queries. The gate turns the reranker from a liability into an asset.
2. **Score-based fusion is fragile.** Any change to BM25 parameters, embedding models, or reranker models can break carefully calibrated score interpolation. RRF is robust to all of these.
3. **Arrow/JS type mismatches are silent.** The vector type bug produced no errors—`NaN` values propagated silently through the pipeline. Strong type validation at storage boundaries is essential.
4. **Benchmark iteratively.** The 12-phase remediation cycle was driven by quantitative benchmarking at each step. Without BEIR evaluation, most of these bugs would remain undetected in production.

9 Limitations and Future Work

- **In-domain evaluation gap**: BEIR evaluates zero-shot cross-domain retrieval. Production data is agent workspace content. A custom agentic evaluation dataset (using LLM-generated relevance judgments) is planned.
- **Single-node**: All inference runs on one DGX Spark. Horizontal scaling would require embedding service load balancing.
- **Gate threshold generalization**: τ_{high} and τ_{low} were tuned on SciFact. Cross-domain generalization needs verification on FiQA and NFCorpus (benchmark in progress).
- **End-to-end evaluation**: Current metrics are retrieval-only. RAGAS [Es et al., 2024] integration for faithfulness and answer quality is planned.
- **Reranker model size**: The 1B-parameter reranker shows score compression issues. A larger model (e.g., 8B) may provide better discrimination at the cost of latency.

10 Conclusion

memory-spark demonstrates that production agentic RAG benefits significantly from adaptive retrieval strategies. The Dynamic Reranker Gate reduces cross-encoder invocations by 78% while simultaneously improving recall—a counter-intuitive result explained by the gate's ability to preserve confident vector rankings that the reranker would otherwise disrupt. Combined with scale-invariant RRF and a 15-stage pipeline engineered through iterative BEIR benchmarking, the system achieves NDCG@10 of 0.7802 on SciFact with sub-second latency, serving 6 concurrent agents in production.

Reproducibility. All code, benchmarks, and evaluation scripts are open-source at https://github.com/kleinpanic/memory-spark.

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